

MACHINE LEARNING FOR SIGNAL PROCESSING

17-2-2025

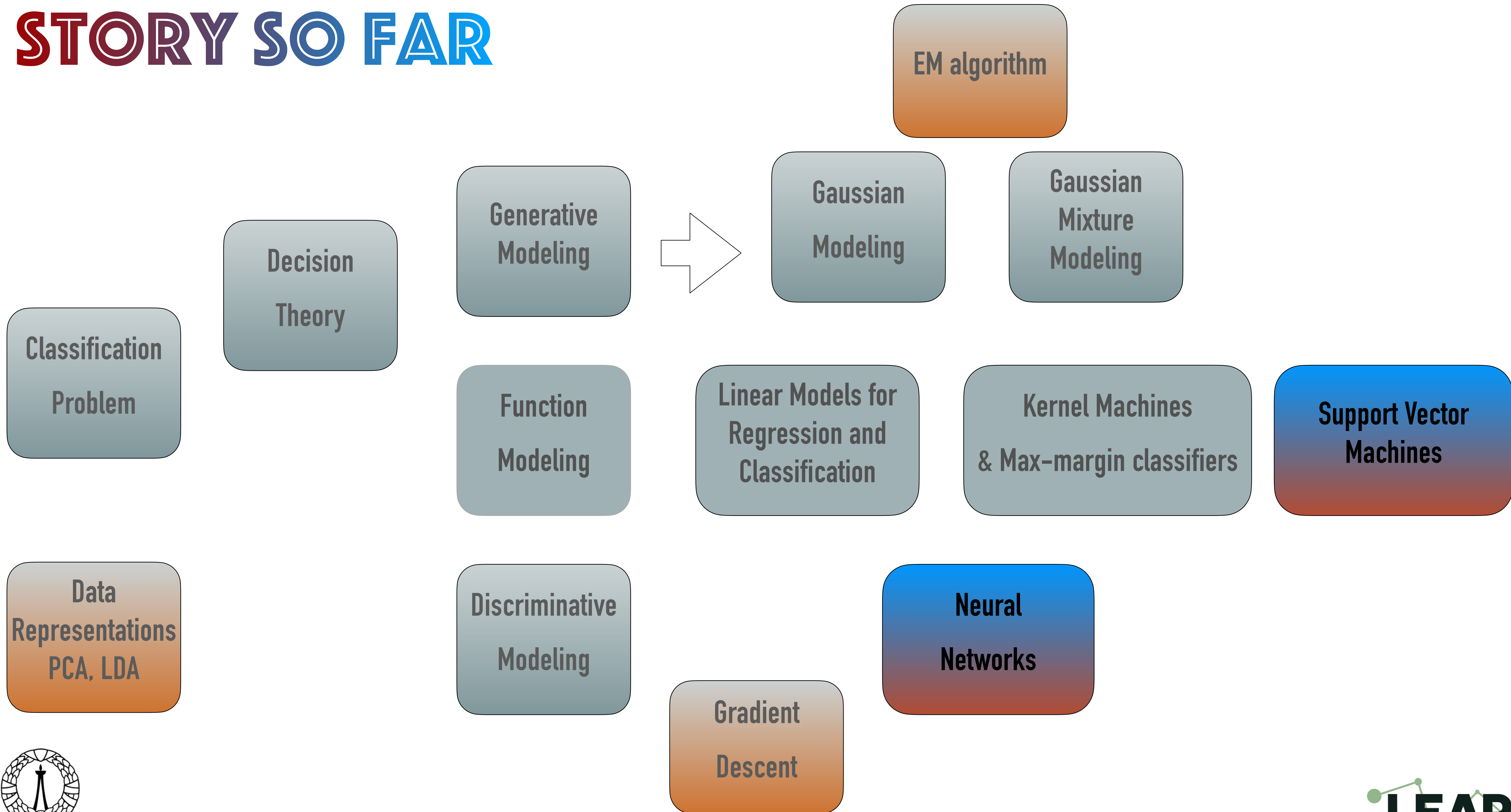
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<http://leap.ee.iisc.ac.in/sriram/teaching/MLSP25/>



STORY SO FAR



EXAMPLES OF KERNEL FUNCTIONS

- Linear: $k(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^T \mathbf{x}_j$
- Polynomial of power p : $k(\mathbf{x}_i, \mathbf{x}_j) = (1 + \mathbf{x}_i^T \mathbf{x}_j)^p$
- Gaussian (radial-basis function network):

$$k(\mathbf{x}_i, \mathbf{x}_j) = \exp \frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{\sigma^2}$$

- Sigmoid: $k(\mathbf{x}_i, \mathbf{x}_j) = \tanh(\beta_0 \mathbf{x}_i^T \mathbf{x}_j + \beta_1)$

PROPERTIES OF KERNEL FUNCTIONS

$$k(\mathbf{x}, \mathbf{x}') = ck_1(\mathbf{x}, \mathbf{x}')$$

$$k(\mathbf{x}, \mathbf{x}') = f(\mathbf{x})k_1(\mathbf{x}, \mathbf{x}')f(\mathbf{x}')$$

$$k(\mathbf{x}, \mathbf{x}') = q(k_1(\mathbf{x}, \mathbf{x}'))$$

$$k(\mathbf{x}, \mathbf{x}') = \exp(k_1(\mathbf{x}, \mathbf{x}'))$$

$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}') + k_2(\mathbf{x}, \mathbf{x}')$$

$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}')k_2(\mathbf{x}, \mathbf{x}')$$

$$k(\mathbf{x}, \mathbf{x}') = k_3(\phi(\mathbf{x}), \phi(\mathbf{x}'))$$

$$k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^T \mathbf{A} \mathbf{x}'$$

$$k(\mathbf{x}, \mathbf{x}') = k_a(\mathbf{x}_a, \mathbf{x}'_a) + k_b(\mathbf{x}_b, \mathbf{x}'_b)$$

$$k(\mathbf{x}, \mathbf{x}') = k_a(\mathbf{x}_a, \mathbf{x}'_a)k_b(\mathbf{x}_b, \mathbf{x}'_b)$$

MAX-MARGIN CLASSIFIERS

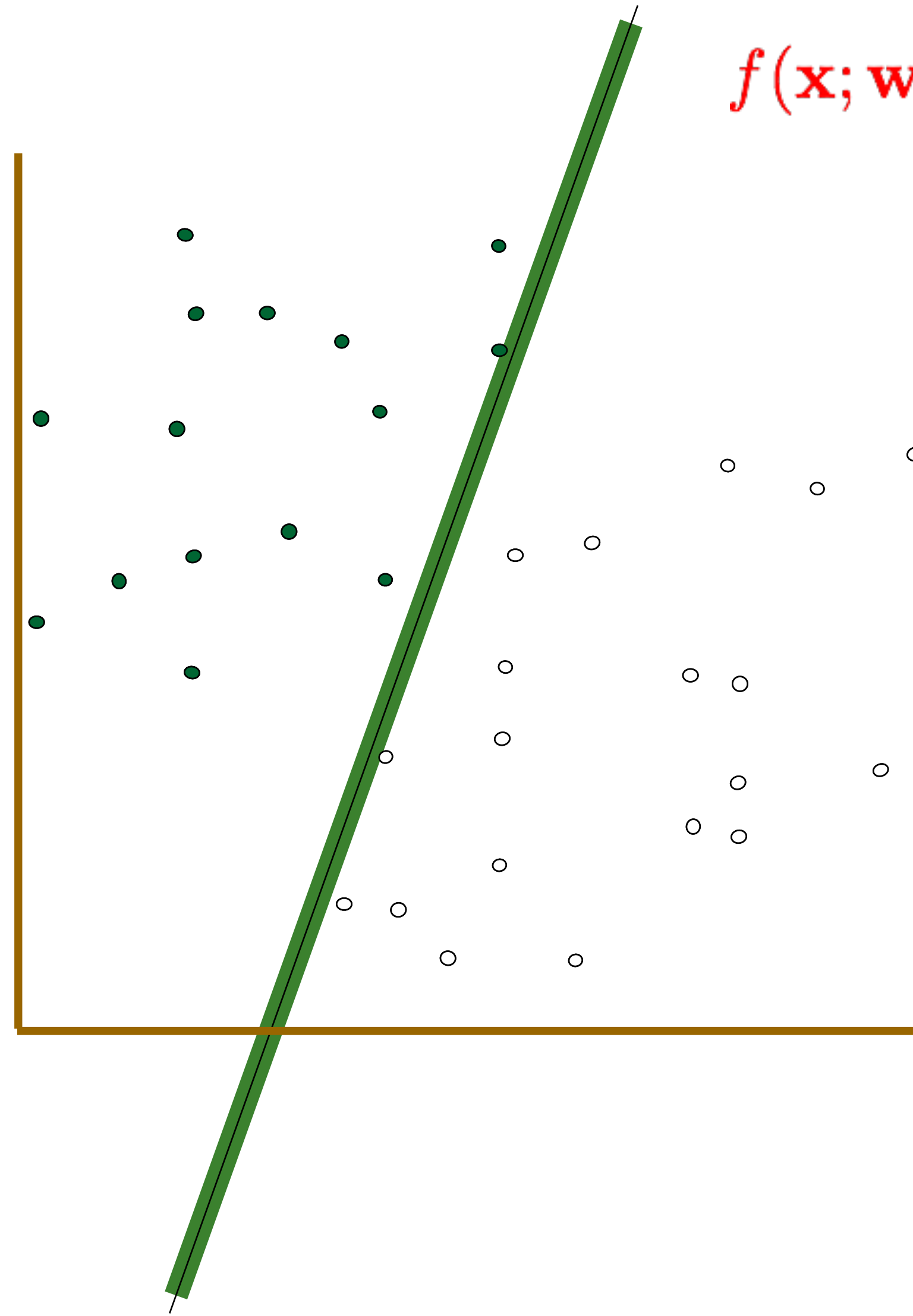
$\mathbf{x}$

f

y_{est}

$$f(\mathbf{x}; \mathbf{w}, b) = \text{sgn}(\mathbf{w}^T \phi(\mathbf{x}) + b)$$

- denotes +1
- denotes -1



Define the **margin** of a linear classifier as the width that the boundary could be increased by before hitting a datapoint.

SVM Formulation

- ❖ Goal - 1) Correctly classify all training data

$$\left. \begin{array}{l} \mathbf{w}^T \phi(\mathbf{x}_n) + b \geq 1 \quad \text{if } t_n = +1 \\ \mathbf{w}^T \phi(\mathbf{x}_n) + b \leq -1 \quad \text{if } t_n = -1 \end{array} \right\} \rightarrow$$
$$t_n(\mathbf{w}^T \phi(\mathbf{x}_n) + b) \geq 1$$

- 2) Define the Margin

$$\frac{1}{\|\mathbf{w}\|} \min_n [t_n(\mathbf{w}^T \phi(\mathbf{x}_n) + b)]$$

- 3) Maximize the Margin

$$\operatorname{argmax}_{\mathbf{w}, b} \left\{ \frac{1}{\|\mathbf{w}\|} \min_n [t_n(\mathbf{w}^T \phi(\mathbf{x}_n) + b)] \right\}$$

- ❖ Equivalently written as

$$\operatorname{argmin}_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 \text{ such that } t_n(\mathbf{w}^T \phi(\mathbf{x}_n) + b) \geq 1$$

Solving the Optimization Problem

- Need to optimize a *quadratic* function subject to *linear* constraints.
- Quadratic optimization problems are a well-known class of mathematical programming problems, and many (rather intricate) algorithms exist for solving them.
- The solution involves constructing a *dual problem* where a *Lagrange multiplier* a_n is associated with every constraint in the primary problem:
- The dual problem in this case is maximized

Find $\{a_1, \dots, a_N\}$ such that

$$\tilde{L}(\mathbf{a}) = \sum_{n=1}^N a_n - \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N t_n t_m a_n a_m k(\mathbf{x}_n, \mathbf{x}_m) \text{ maximized}$$

and $\sum_n a_n t_n = 0 \quad a_n \geq 0$

Solving the Optimization Problem

- The solution has the form:

$$\mathbf{w} = \sum_{n=1}^N a_n \phi(\mathbf{x}_n)$$

- Each non-zero a_n indicates that corresponding $\mathbf{x}_n$ is a support vector. Let S denote the set of support vectors.

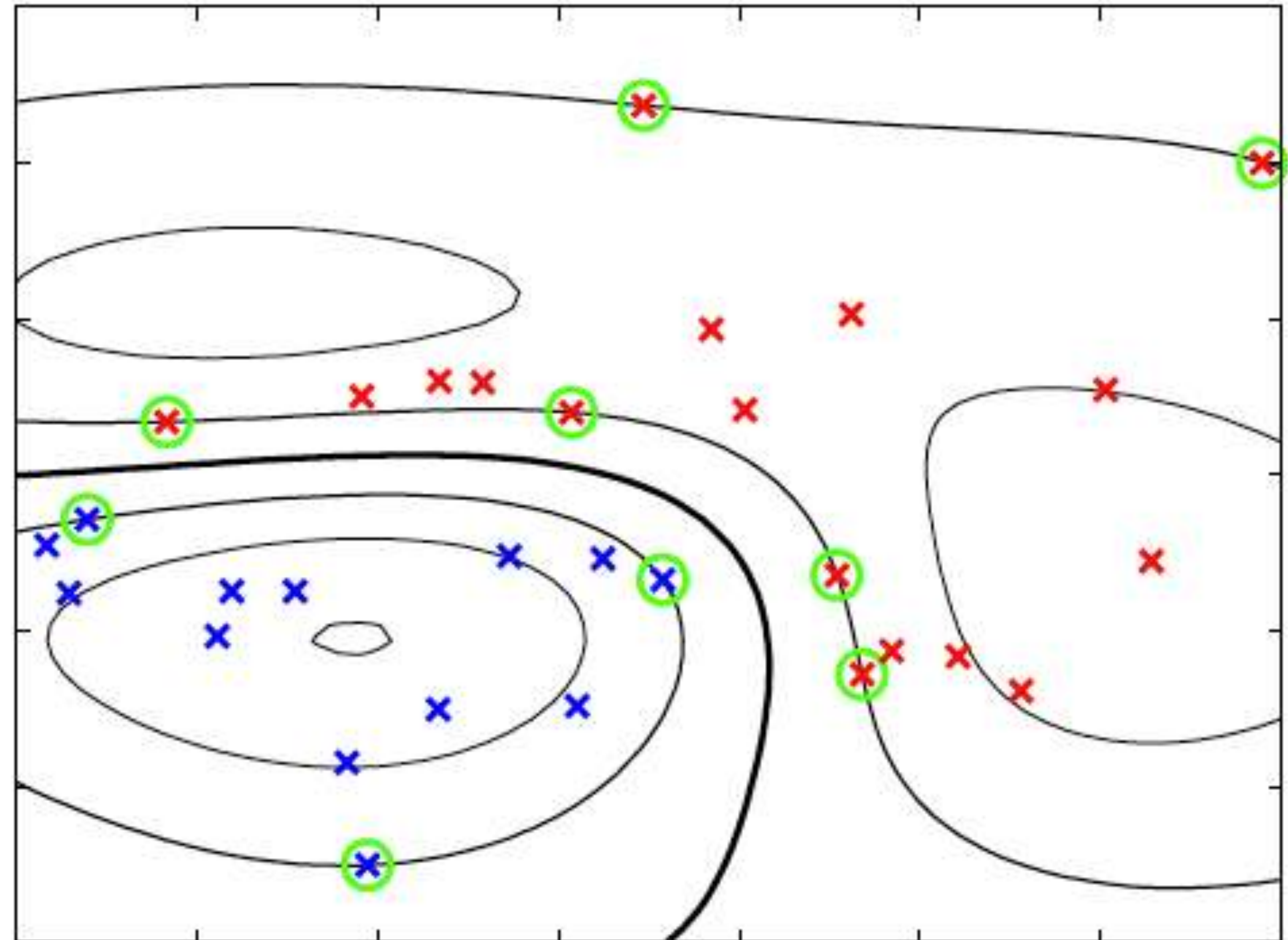
$$b = y(\mathbf{x}_n) - \sum_{m \in S} a_m k(\mathbf{x}_m, \mathbf{x}_n)$$

- And the classifying function will have the form:

$$y(\mathbf{x}) = \sum_{n \in S} a_n k(\mathbf{x}_n, \mathbf{x}) + b$$

GAUSSIAN KERNEL VISUALIZATION

Example of synthetic data from two classes in two dimensions showing contours of constant $y(\mathbf{x})$ obtained from a support vector machine having a Gaussian kernel function. Also shown are the decision boundary, the margin boundaries, and the support vectors.



Overlapping class boundaries

- The classes are not linearly separable - Introducing slack variables ζ_n
- Slack variables are non-negative $\zeta_n \geq 0$
- They are defined using

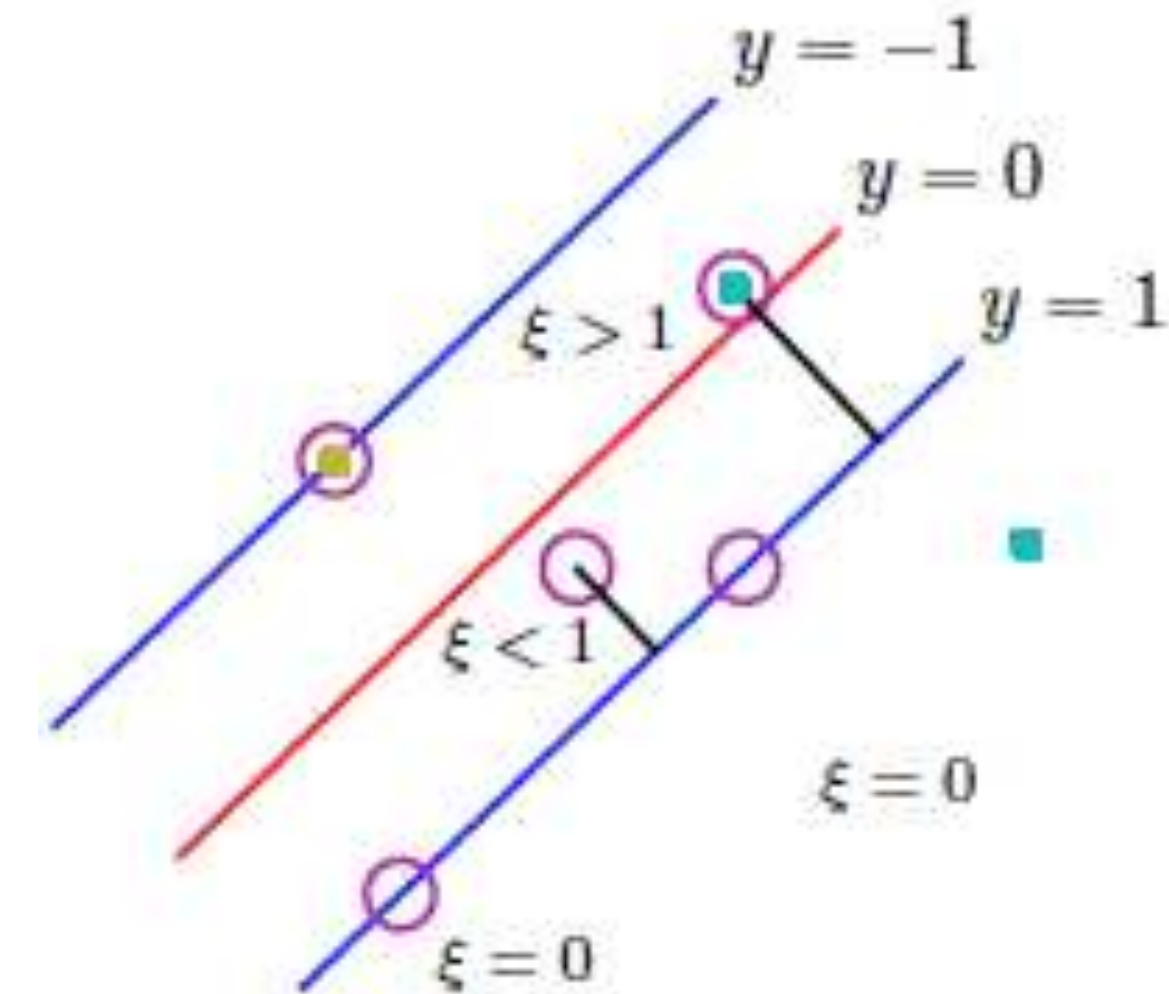
$$t_n y(\mathbf{x}_n) \geq 1 - \zeta_n$$

- The upper bound on mis-classification

$$\sum_n \zeta_n$$

- The cost function to be optimized in this case

$$C \sum_n \zeta_n + \frac{1}{2} \mathbf{w}^T \mathbf{w}$$



SVM Formulation - overlapping classes

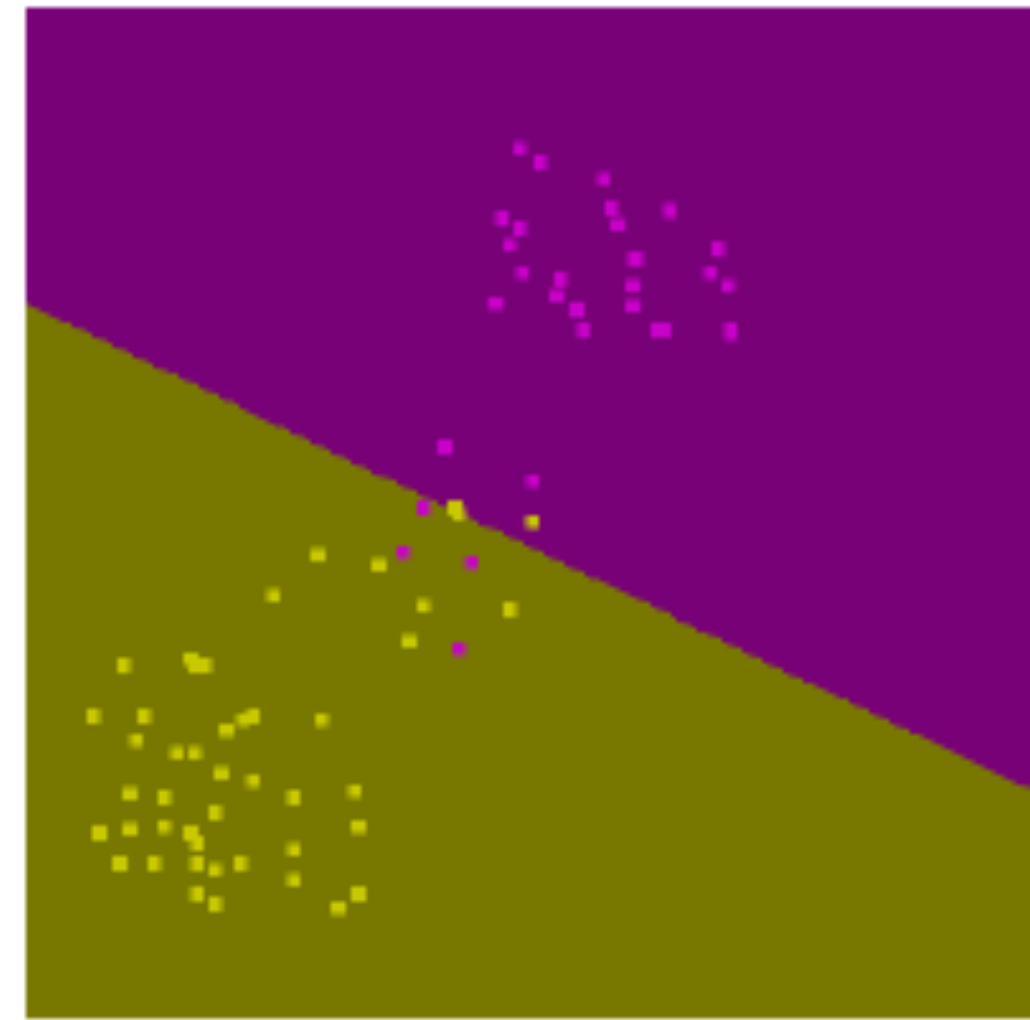
- Formulation very similar to previous case except for additional constraints

$$0 \leq a_n \leq C$$

- Solved using the dual formulation - sequential minimal optimization algorithm
- Final classifier is based on the sign of

$$y(\mathbf{x}) = \sum_{n \in S} a_n k(\mathbf{x}_n, \mathbf{x}) + b$$

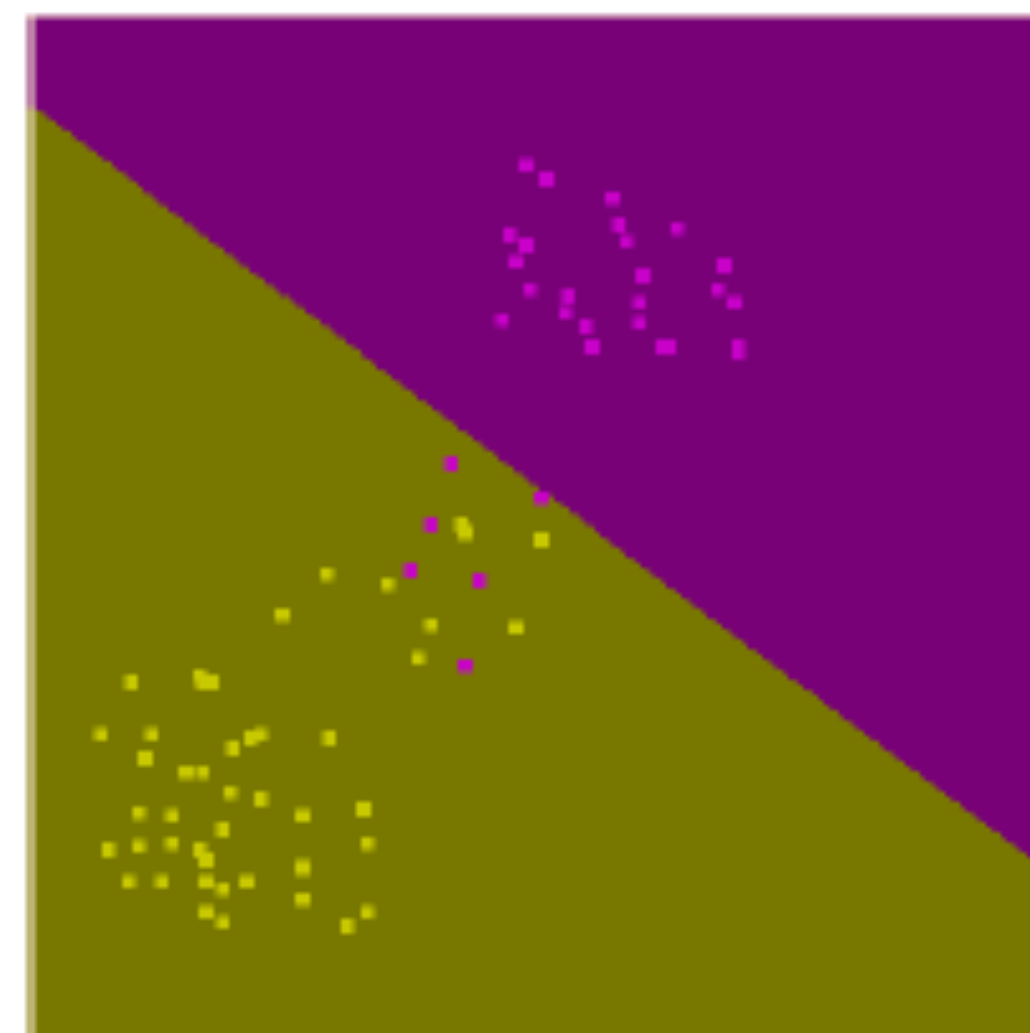
Overlapping class boundaries



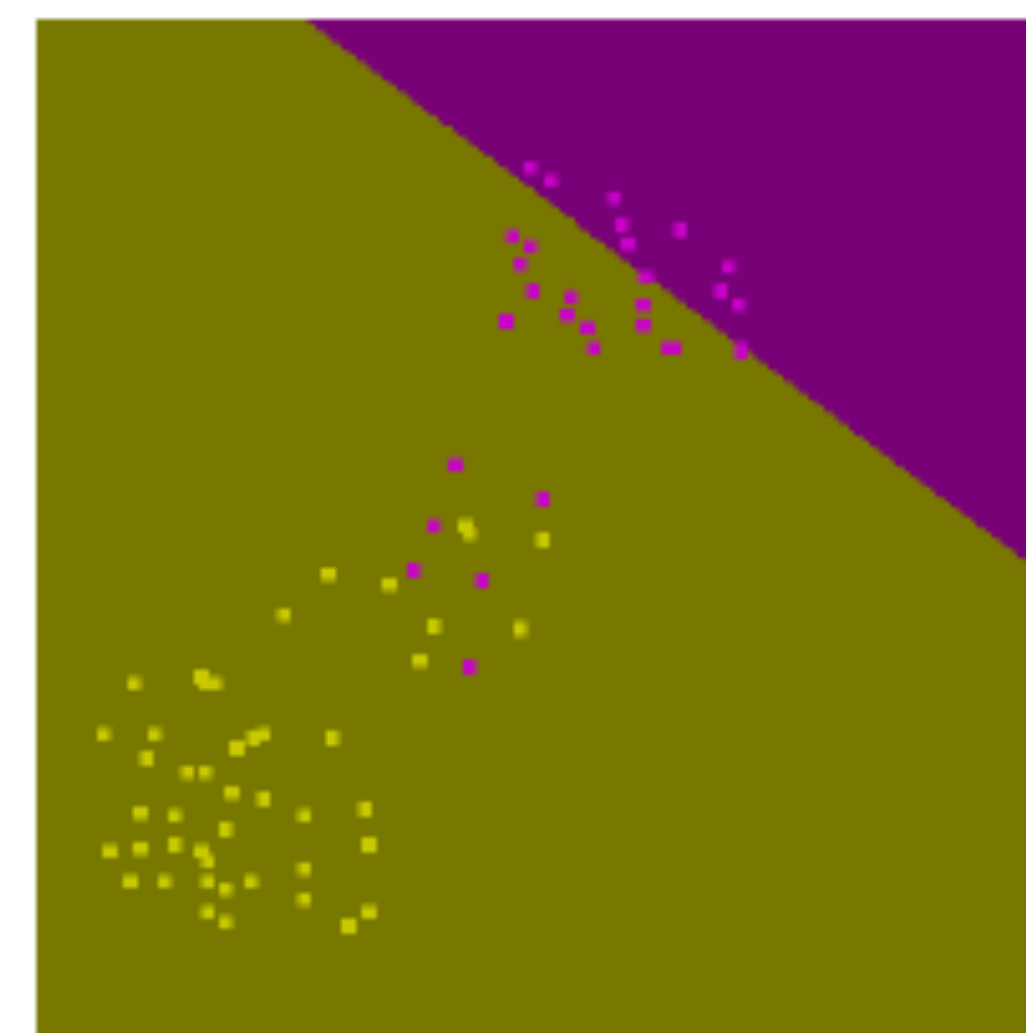
$C=100$



$C=1$



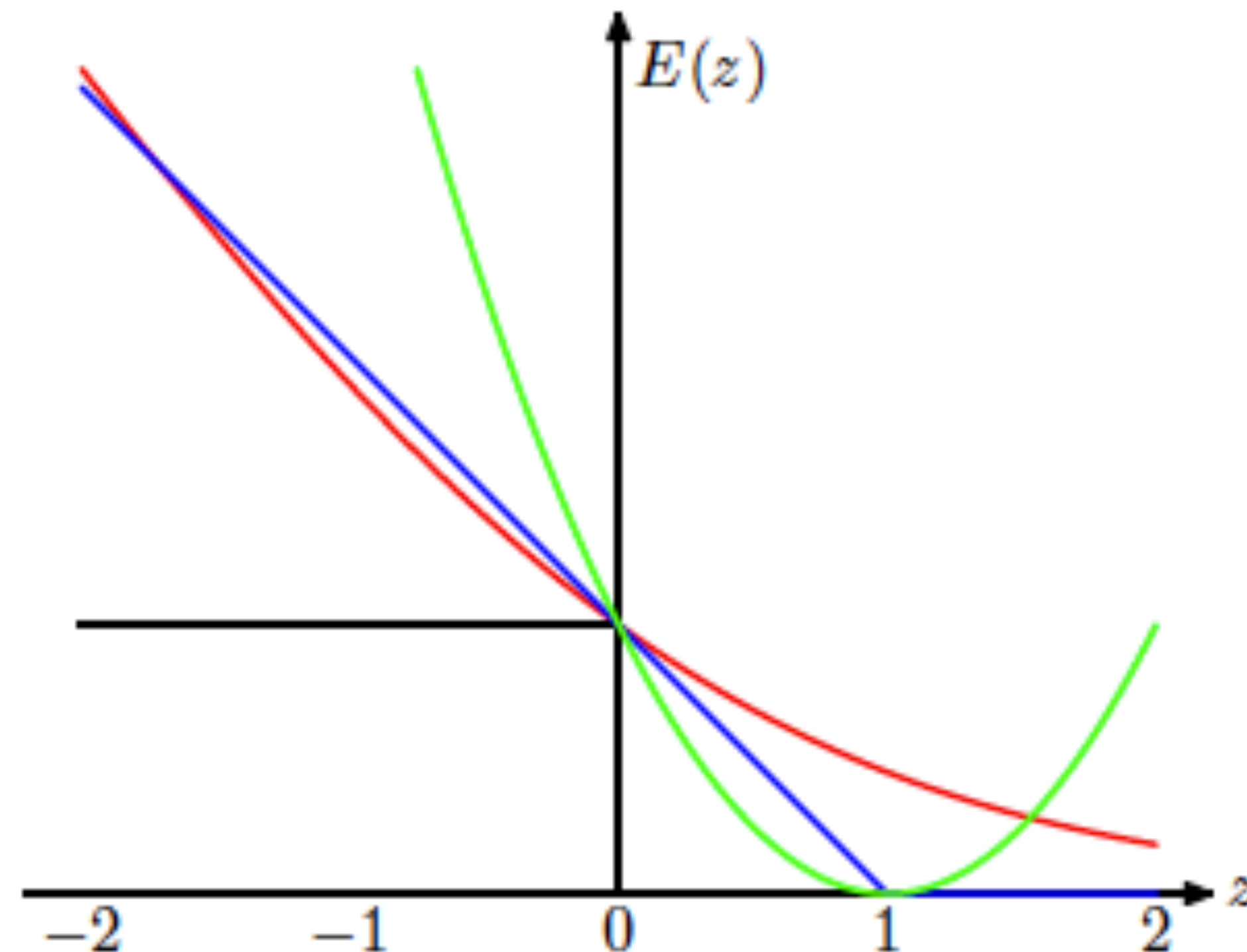
$C=0.15$



$C=0.1$

CONNECTION WITH OTHER MODELS

Plot of the 'hinge' error function used in support vector machines, shown in blue, along with the error function for logistic regression, rescaled by a factor of $1/\ln(2)$ so that it passes through the point $(0, 1)$, shown in red. Also shown are the misclassification error in black and the squared error in green.



Properties of SVM

- Flexibility in choosing a similarity function
- **Sparseness** of solution when dealing with large data sets
 - only support vectors are used to specify the separating hyperplane
- Ability to **handle large feature spaces**
 - complexity does not depend on the dimensionality of the feature space
- Overfitting can be controlled by soft margin approach
- **Nice math property**: a simple convex optimization problem which is guaranteed to converge to a single global solution
- Feature Selection

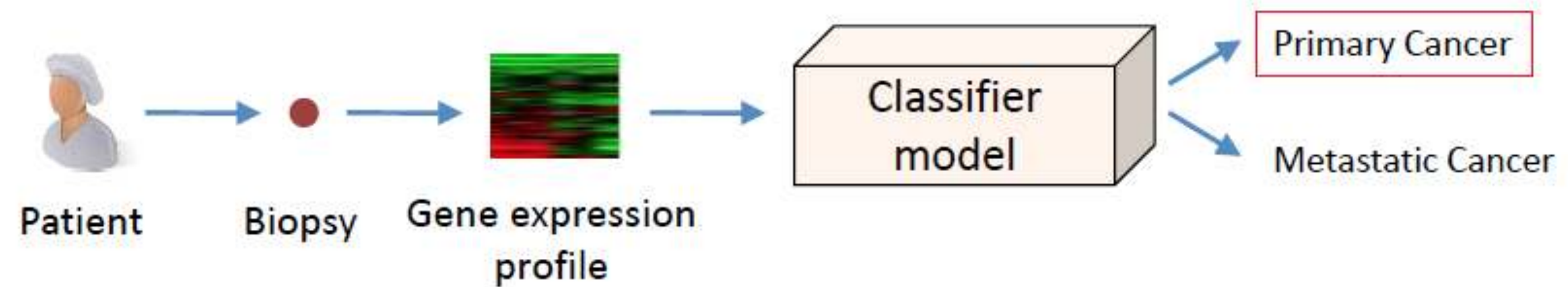
SVM Applications

- SVM has been used successfully in many real-world problems
 - text (and hypertext) categorization
 - image classification
 - bioinformatics (Protein classification, Cancer classification)
 - hand-written character recognition

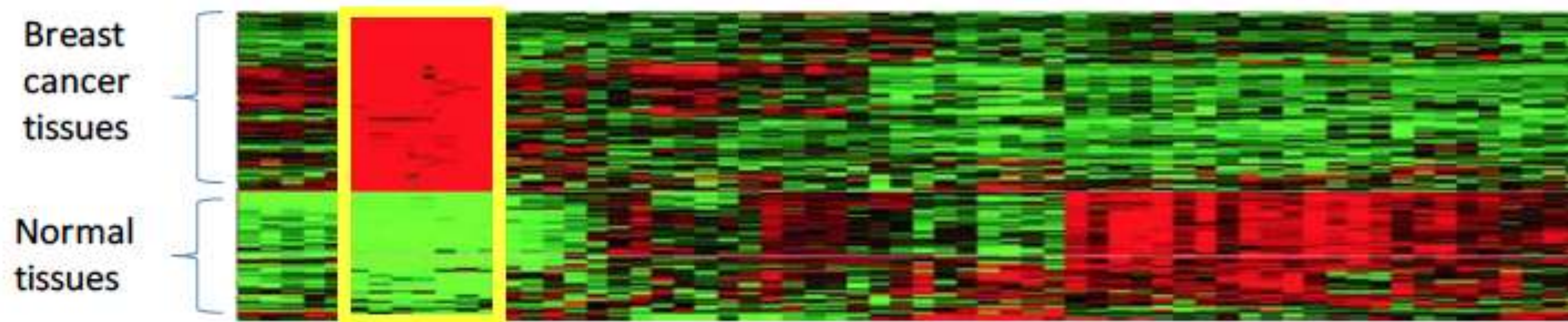
Application 1: Cancer Classification

- SVM has been used successfully in many real-world problems

- text (and hypertext) categorization
- image classification
- bioinformatics (Protein classification, Cancer classification)
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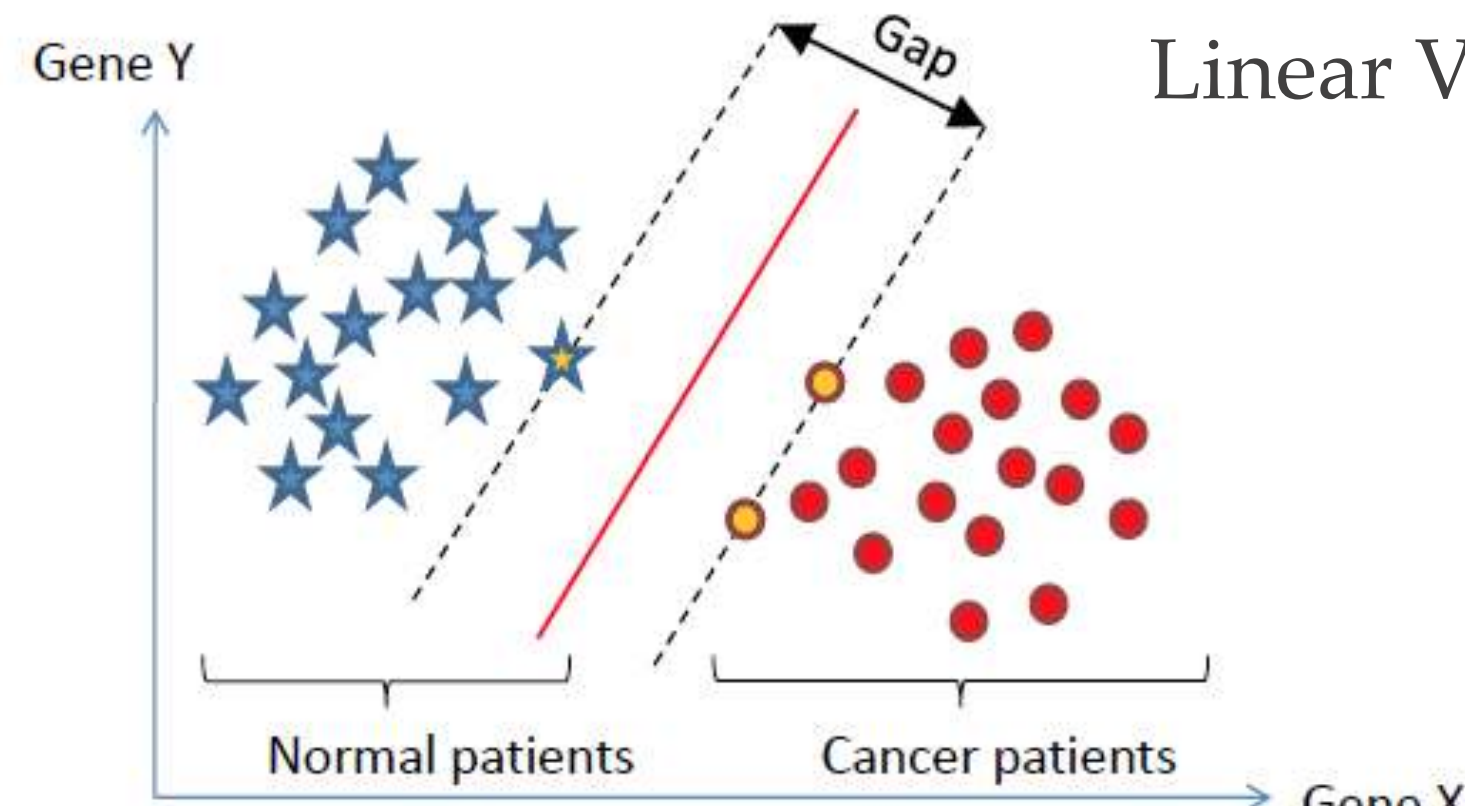


- E.g., find the most compact panel of breast cancer biomarkers from microarray gene expression data for 20,000 genes:

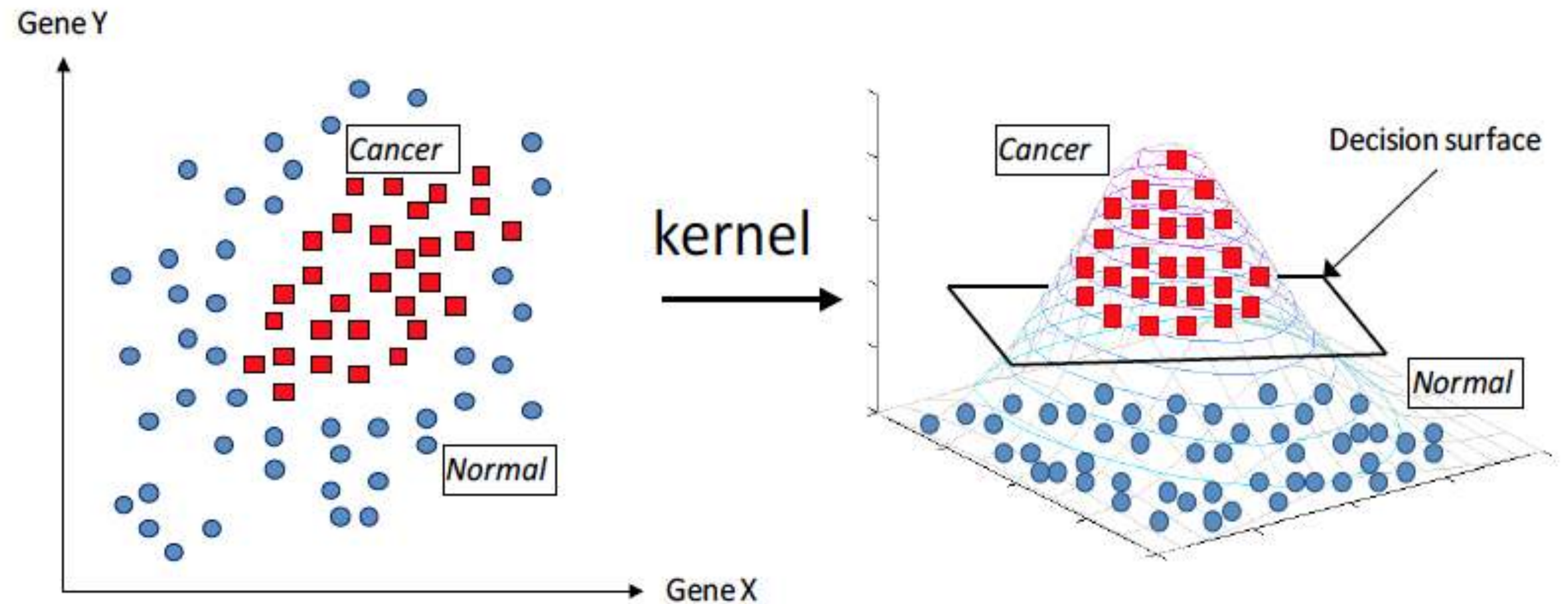


"Gentle Introduction to SVMs in Biomedicine", Statnikov et al. NYU school of medicine

Application 1: Cancer Classification



Linear Versus Non-linear SVMs



Weakness of SVM

- It is sensitive to noise
 - A relatively small number of mislabeled examples can dramatically decrease the performance
- It only considers two classes
 - how to do multi-class classification with SVM?
 - Answer:
 - 1) with output m , learn m SVM's
 - SVM 1 learns "Output==1" vs "Output != 1"
 - SVM 2 learns "Output==2" vs "Output != 2"
 - :
 - SVM m learns "Output== m " vs "Output != m "
 - 2) To predict the output for a new input, just predict with each SVM and find out which one puts the prediction the furthest into the positive region.

Application 2: Text Categorization

- Task: The classification of natural text (or hypertext) documents into a fixed number of predefined categories based on their content.
 - email filtering, web searching, sorting documents by topic, etc..
- A document can be assigned to more than one category, so this can be viewed as a series of binary classification problems, one for each category.

Application 2: Text Categorization

IR's vector space model (aka bag-of-words representation)

- A doc is represented by a vector indexed by a pre-fixed set or dictionary of terms
- Values of an entry can be binary or weights

$$\phi_i(x) = \frac{\text{tf}_i \log(\text{idf}_i)}{\kappa},$$

- Doc $x \Rightarrow \phi(x)$

Application 3: Handwriting Recognition

For example MNIST hand-writing recognition.
60,000 training examples, 10000 test examples, 28x28.

Linear SVM has around 8.5% test error.

Polynomial SVM has around 1% test error.

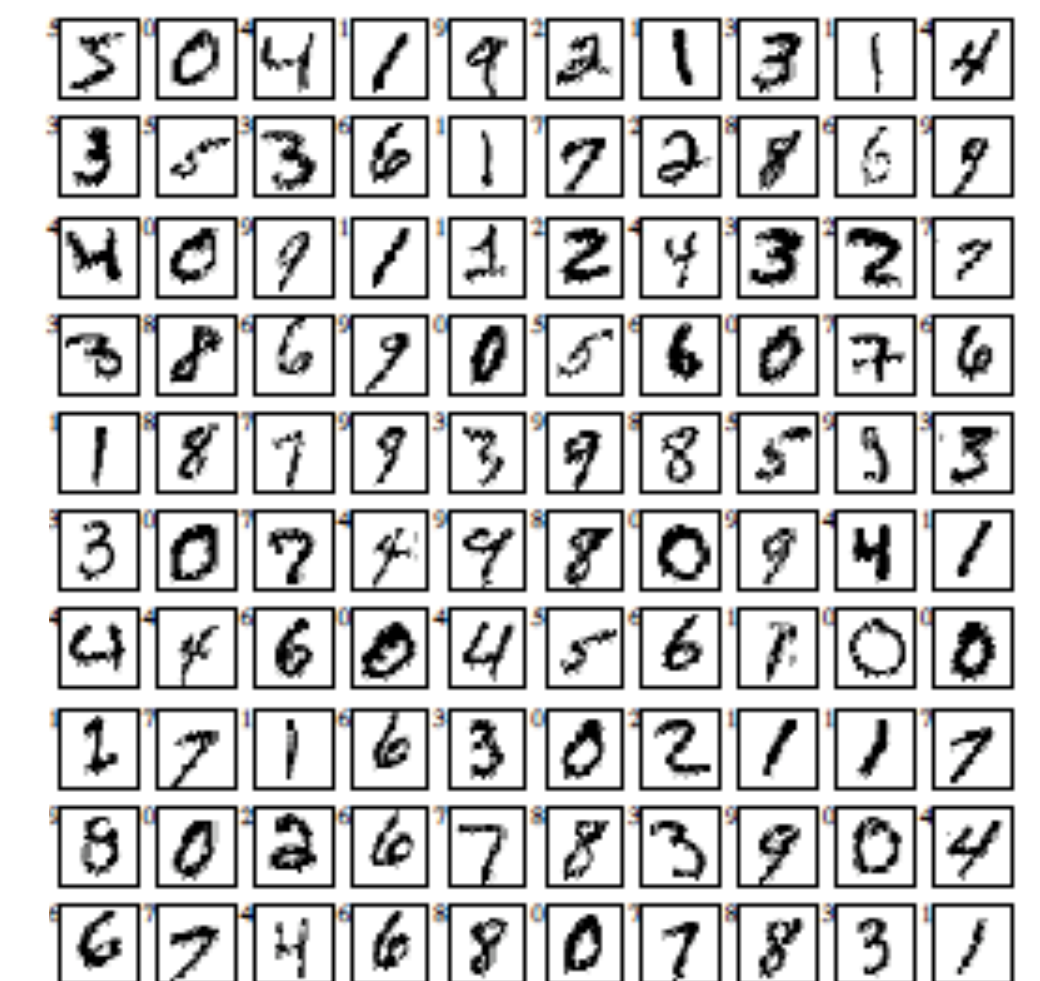
- The distance between two documents is $\langle \phi(x) \phi(z) \rangle$

- $K(x,z) = \langle \phi(x) \phi(z) \rangle$ is a valid kernel, SVM can be used with $K(x,z)$ for discrimination.

23 SVMs : full MNIST results

- Why SVM?
 - High dimensional input space
 - Few irrelevant features (dense concept)
 - Sparse document vectors (sparse instances)
 - Text categorization problems are linearly separable

Classifier	Test Error
linear	8.4%
3-nearest-neighbor	2.4%
RBF-SVM	1.4 %



Some Considerations

- Choice of kernel
 - Gaussian or polynomial kernel is default
 - if ineffective, more elaborate kernels are needed
 - domain experts can give assistance in formulating appropriate similarity measures
- Choice of kernel parameters
 - e.g. σ in Gaussian kernel
 - σ is the distance between closest points with different classifications
 - In the absence of reliable criteria, applications rely on the use of a validation set or cross-validation to set such parameters.
- Optimization criterion – Hard margin v.s. Soft margin
 - a lengthy series of experiments in which various parameters are tested

30 SVMs : software

Lots of SVM software:

- LibSVM (C++)
- SVMLight (C)

As well as complete machine learning toolboxes that include SVMs:

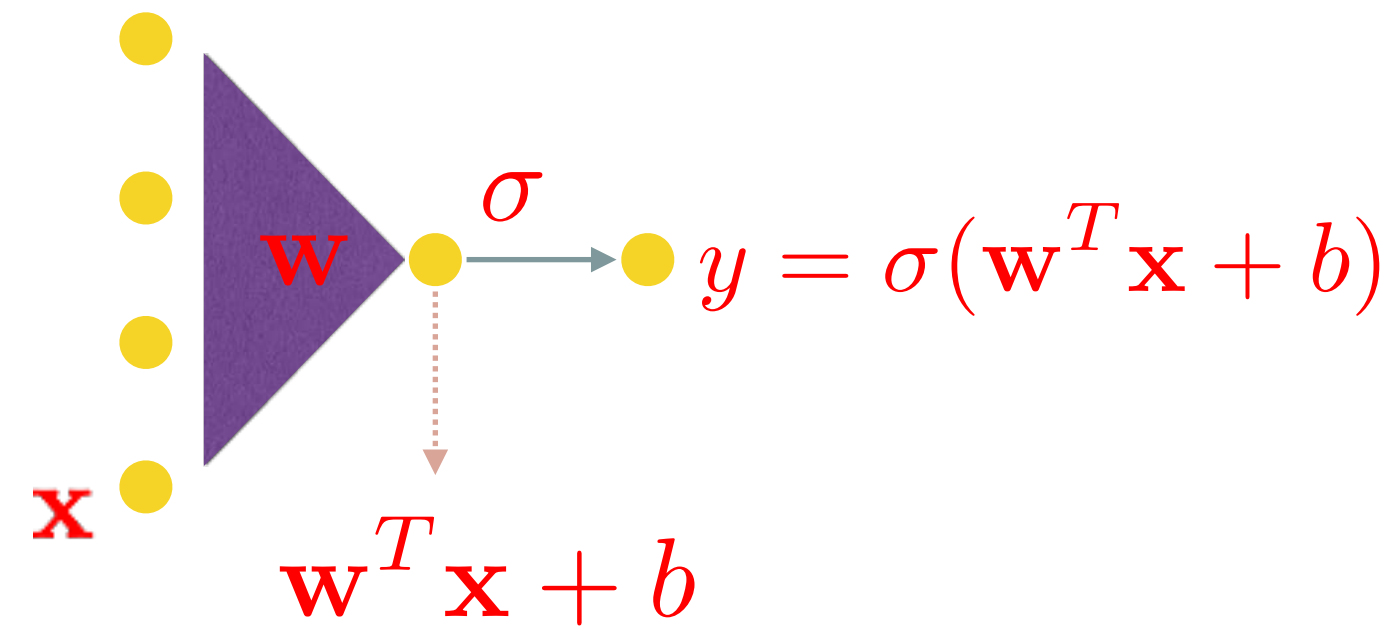
- Torch (C++)
- Spider (Matlab)
- Weka (Java)

All available through www.kernel-machines.org.

NEURAL NETWORKS AND DEEP LEARNING

VISUALIZING LOGISTIC REGRESSION AS A NEURAL NETWORK

- ❖ A logistic regression is the simplest neural network
 - Number of parameters in the model - $D+1$



MULTI-CLASS LOGISTIC REGRESSION

- ❖ Targets are one-hot encoded vectors
 - Model approximates class posteriors using
 - softmax function

$$\text{softmax}(\mathbf{a}) = \begin{bmatrix} \frac{e^{a_1}}{\sum_j e^{a_j}} \\ \frac{e^{a_2}}{\sum_j e^{a_j}} \\ \vdots \\ \frac{e^{a_K}}{\sum_j e^{a_j}} \end{bmatrix}$$

$\mathbf{y} = \text{softmax}(\mathbf{W}^T \mathbf{x} + \mathbf{b})$

$\mathbf{W} \in \mathcal{R}^{K \times D} \quad \mathbf{b} \in \mathcal{R}^K$

SOFTMAX FUNCTION

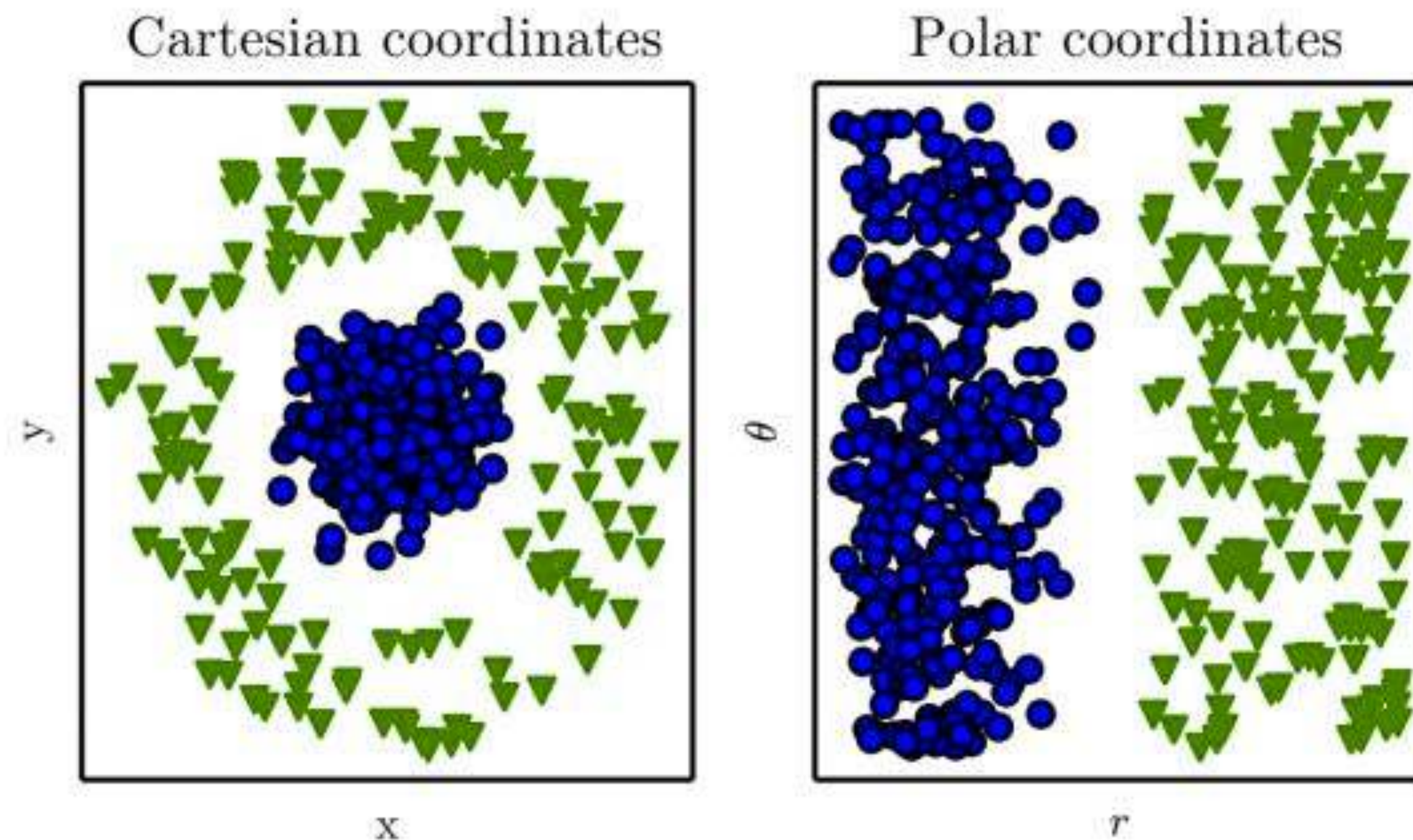
- ❖ Each value is positive
- ❖ Sum of the vector is 1.0
 - Can be interpreted as class posterior probabilities

$$\text{softmax}(\mathbf{a}) = \begin{bmatrix} \frac{e^{a_1}}{\sum_j e^{a_j}} \\ \frac{e^{a_2}}{\sum_j e^{a_j}} \\ \cdot \\ \cdot \\ \frac{e^{a_K}}{\sum_j e^{a_j}} \end{bmatrix}$$

$$\begin{bmatrix} p(C_1|\mathbf{x}) \\ p(C_2|\mathbf{x}) \\ \cdot \\ \cdot \\ p(C_K|\mathbf{x}) \end{bmatrix}$$

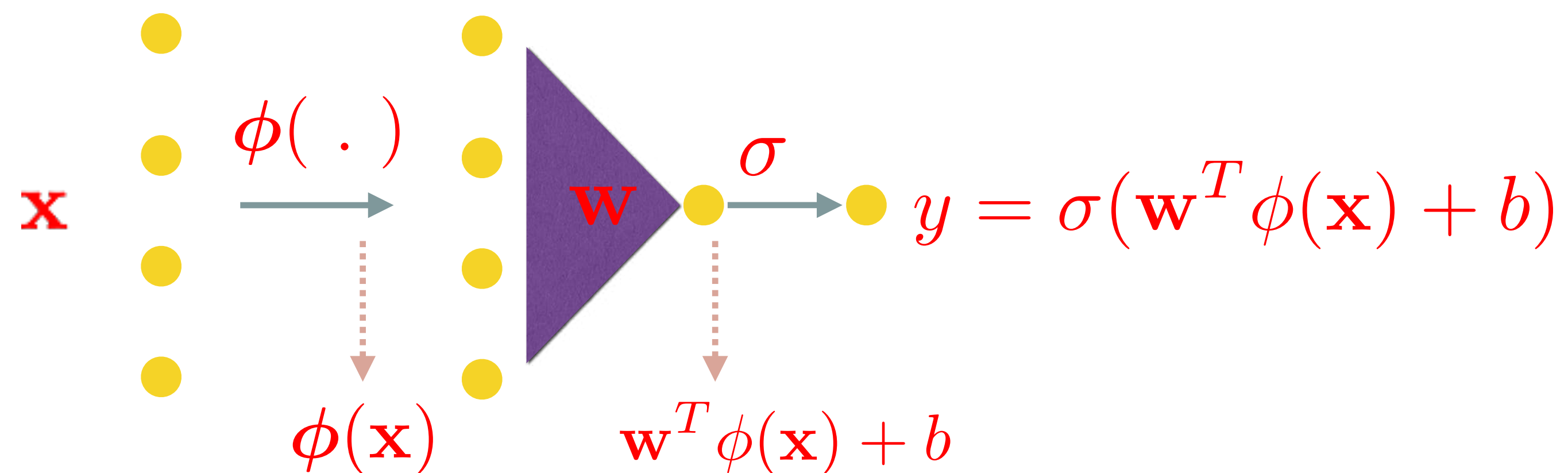
QUESTION

- ❖ **Can we transform the data to linearly separable space**
 - then apply the logistic regression to find the classifier.
 - Example



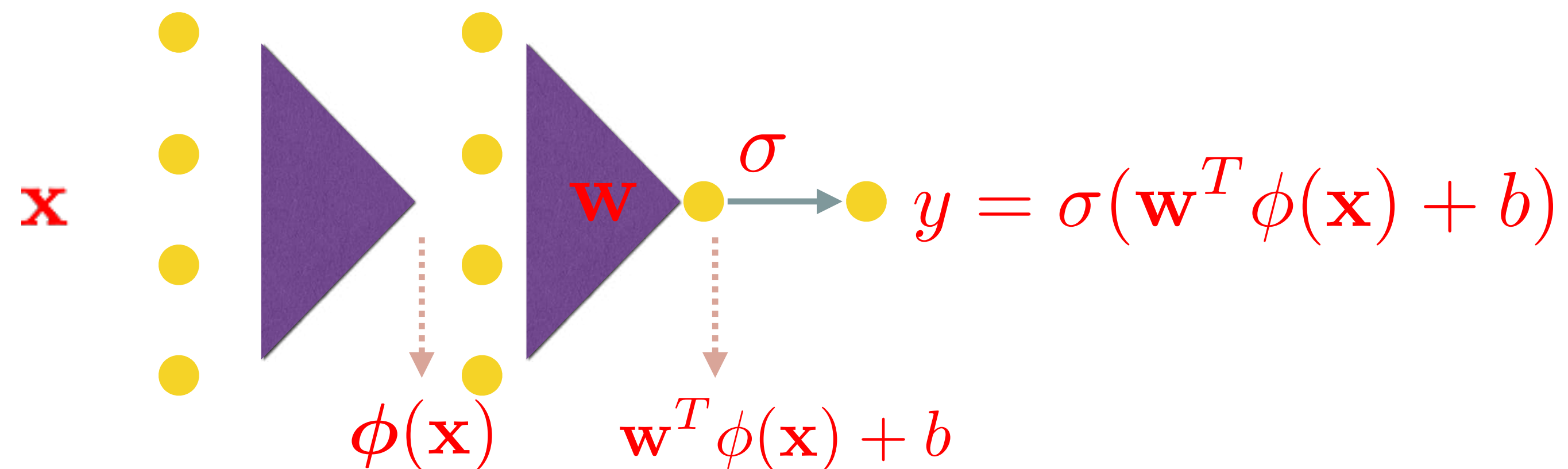
QUESTION

- ❖ **Can we transform the data to linearly separable space**
 - then apply the logistic regression to find the classifier.
 - Can we learn such a transform from the data itself
 - non-linear transformation of the data is needed



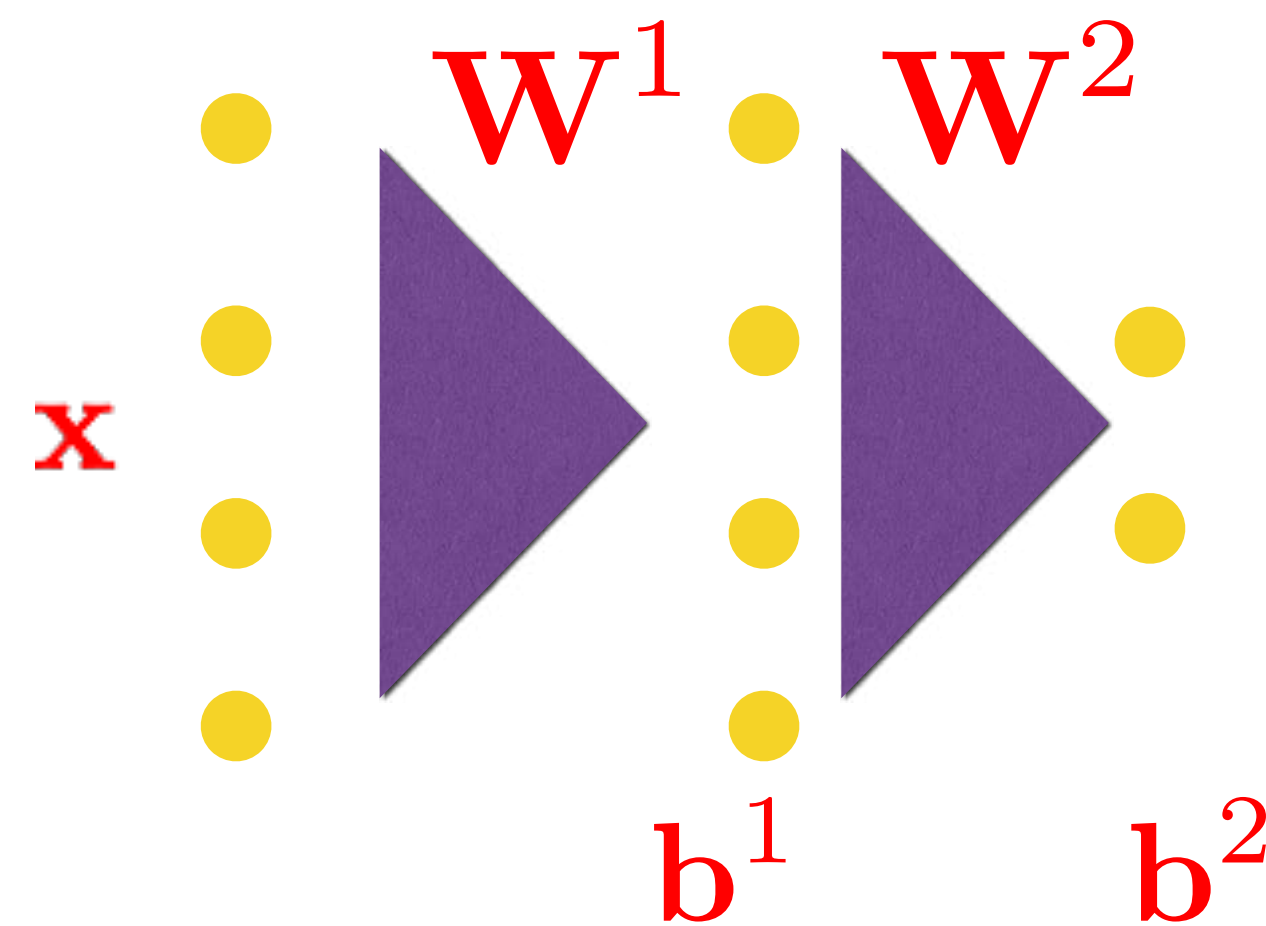
QUESTION

- ❖ **Can we transform the data to linearly separable space**
 - then apply the logistic regression to find the classifier.
- **Can we learn such a transform from the data itself**
 - non-linear transformation of the data is needed.
 - can this also be realized as neural layer



NEURAL NETWORK - 1- HIDDEN LAYER

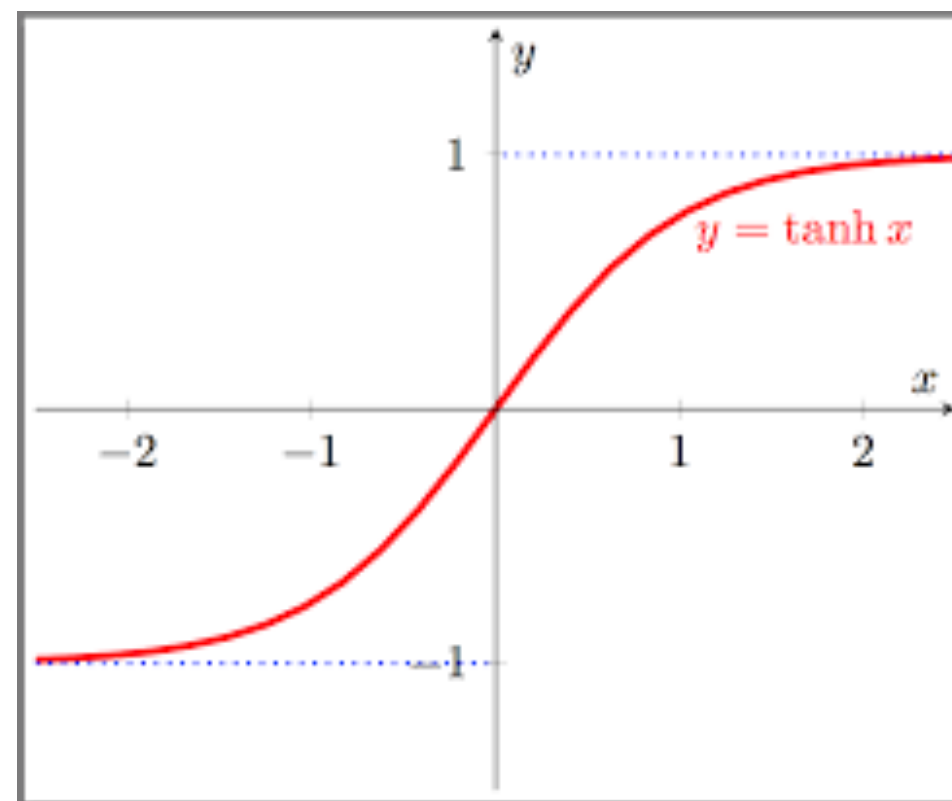
- ❖ Has more capacity than logistic regression
 - can learn non-linear data separation functions
 - both 2-class and K-class classification possible
 - can be learnt using gradient descent



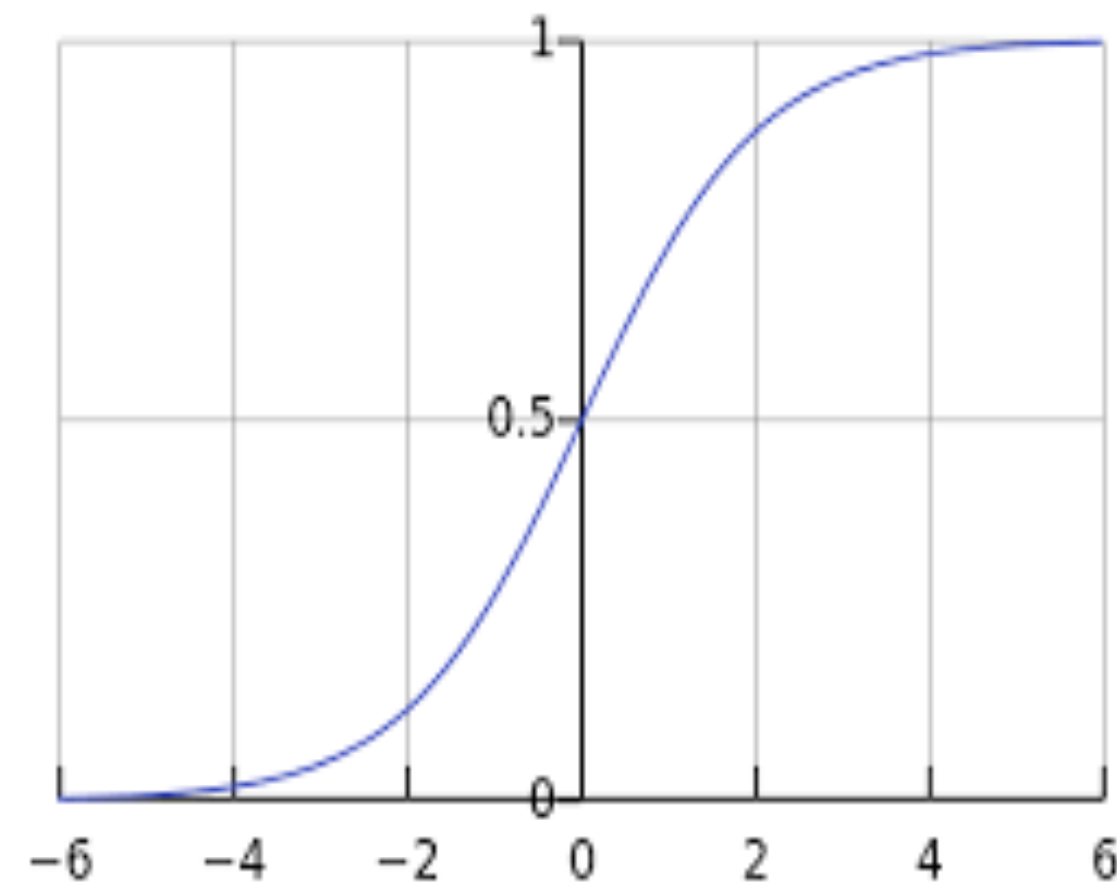
TYPES OF NON-LINEARITIES

Non-linearity in hidden layer

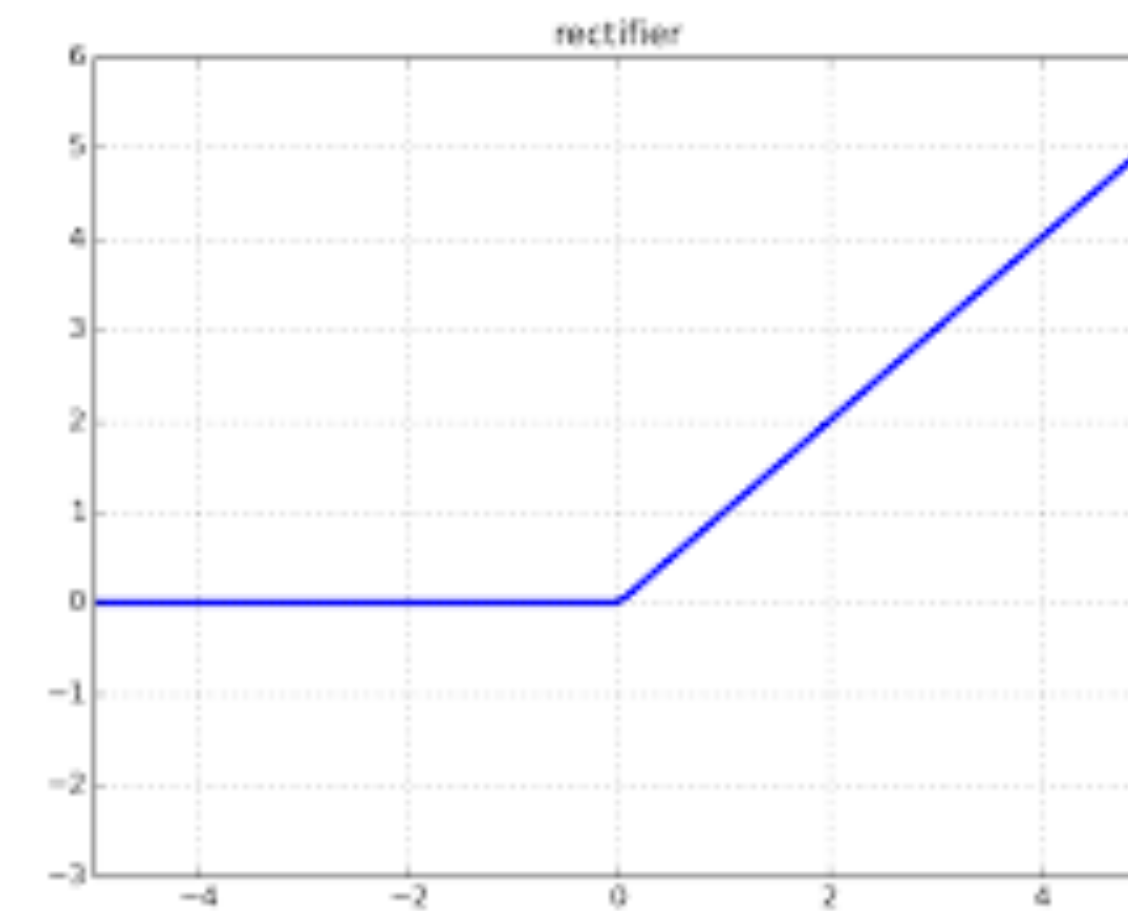
tanh



sigmoid



ReLu



OUTPUT LAYER NON-LINEARITY AND COST FUNCTIONS

- ❖ Using a softmax non-linearity
 - error function is cross entropy

$$E_{CE} = - \sum_n \sum_k t_{nk} \log(v_{nk})$$

- ❖ For regression style tasks - output is linear
 - error function is mean square error

$$E_{MSE} = - \sum_n \sum_k (t_{nk} - v_{nk})^2$$

FORWARD THROUGH THE MODEL PROPAGATION LEARNING

- ❖ Computations in the forward direction

$$\begin{aligned}\mathbf{a}^1 &= \mathbf{W}^1 \mathbf{x} + \mathbf{b}^1 \\ \mathbf{z}^1 &= \sigma(\mathbf{a}^1) \\ \mathbf{a}^2 &= \mathbf{W}^2 \mathbf{z}^1 + \mathbf{b}^2 \\ \mathbf{y} &= \text{softmax}(\mathbf{a}^2)\end{aligned}$$

- ❖ Loss function

$$E_{CE} = - \sum_n \sum_k t_{nk} \log(v_{nk})$$

$$\Theta = \{\mathbf{W}^1, \mathbf{b}^1, \mathbf{W}^2, \mathbf{b}^2\}$$

- ❖ Parameters in the model

Need to be updated based on the gradients w.r.t. the error

GRADIENT COMPUTATION IN THE MODEL

$$\begin{aligned}\mathbf{a}^1 &= \mathbf{W}^1 \mathbf{x} + \mathbf{b}^1 \\ \mathbf{z}^1 &= \sigma(\mathbf{a}^1) \\ \mathbf{a}^2 &= \mathbf{W}^2 \mathbf{z}^1 + \mathbf{b}^2 \\ \mathbf{y} &= \text{softmax}(\mathbf{a}^2)\end{aligned}$$

$$E_{CE} = - \sum_n \sum_k t_{nk} \log(v_{nk})$$

❖ When computing the gradients

➤ Order of computations

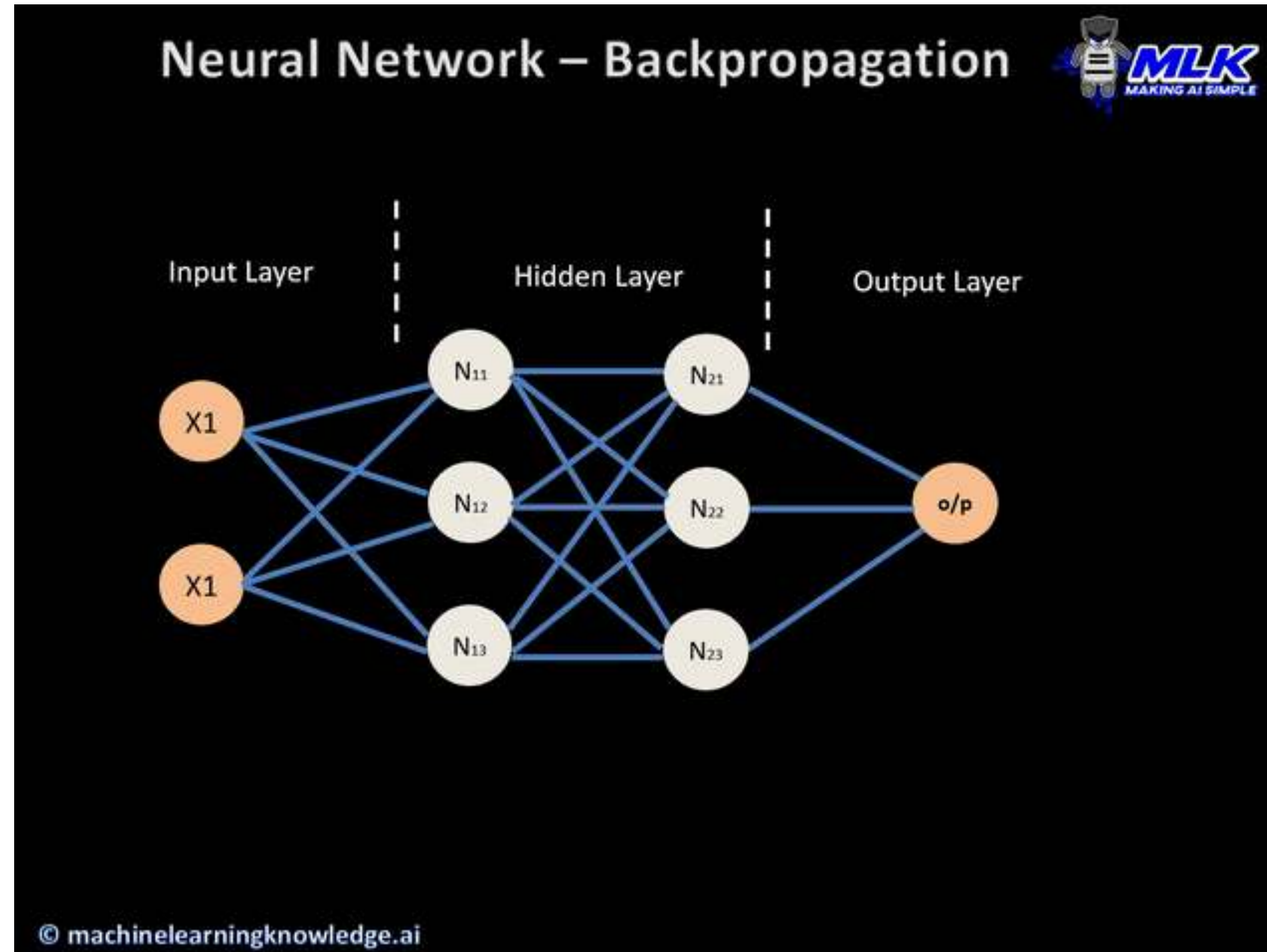
➤ The derivative of the loss function w.r.t output layer

➤ The derivative of the loss function w.r.t output activation

➤ The derivative of the loss function w.r.t hidden layer outputs

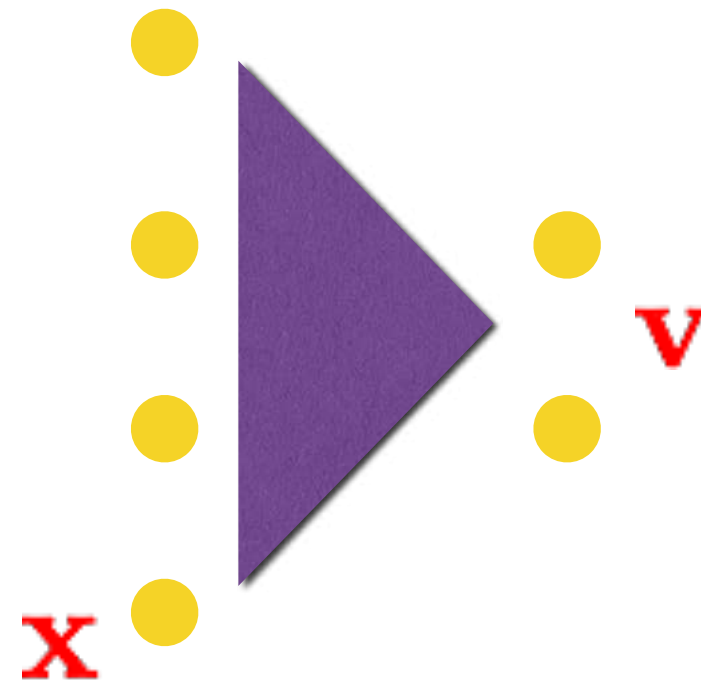
➤ The derivative of the loss function w.r.t. hidden layer activations

BACK PROPAGATION LEARNING



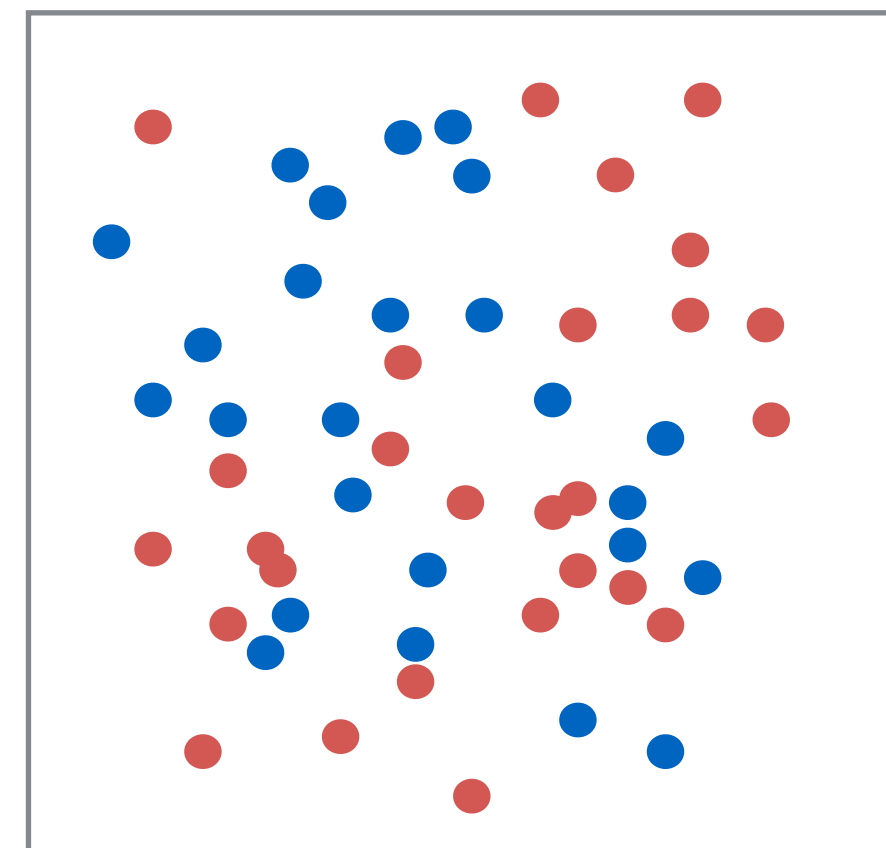
PERCEPTRON ALGORITHM

Perceptron Model [McCulloch, 1943, Rosenblatt, 1957]



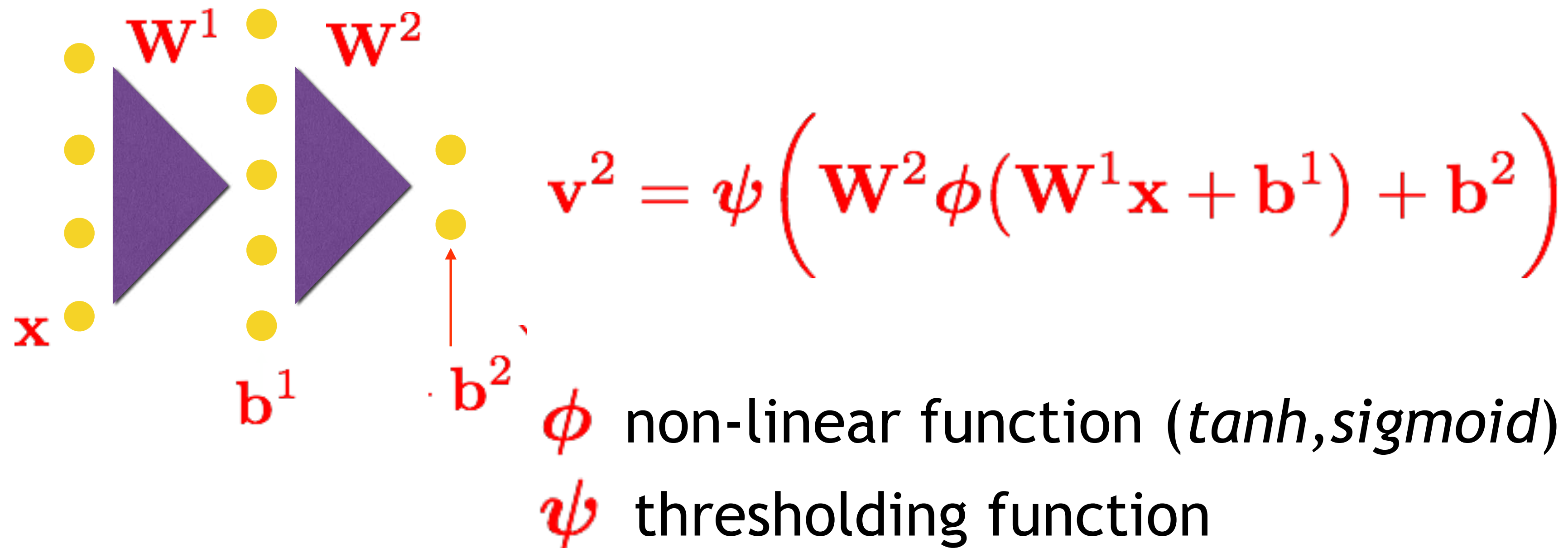
Targets are binary classes $[-1, 1]$

What if the data is not
linearly separable



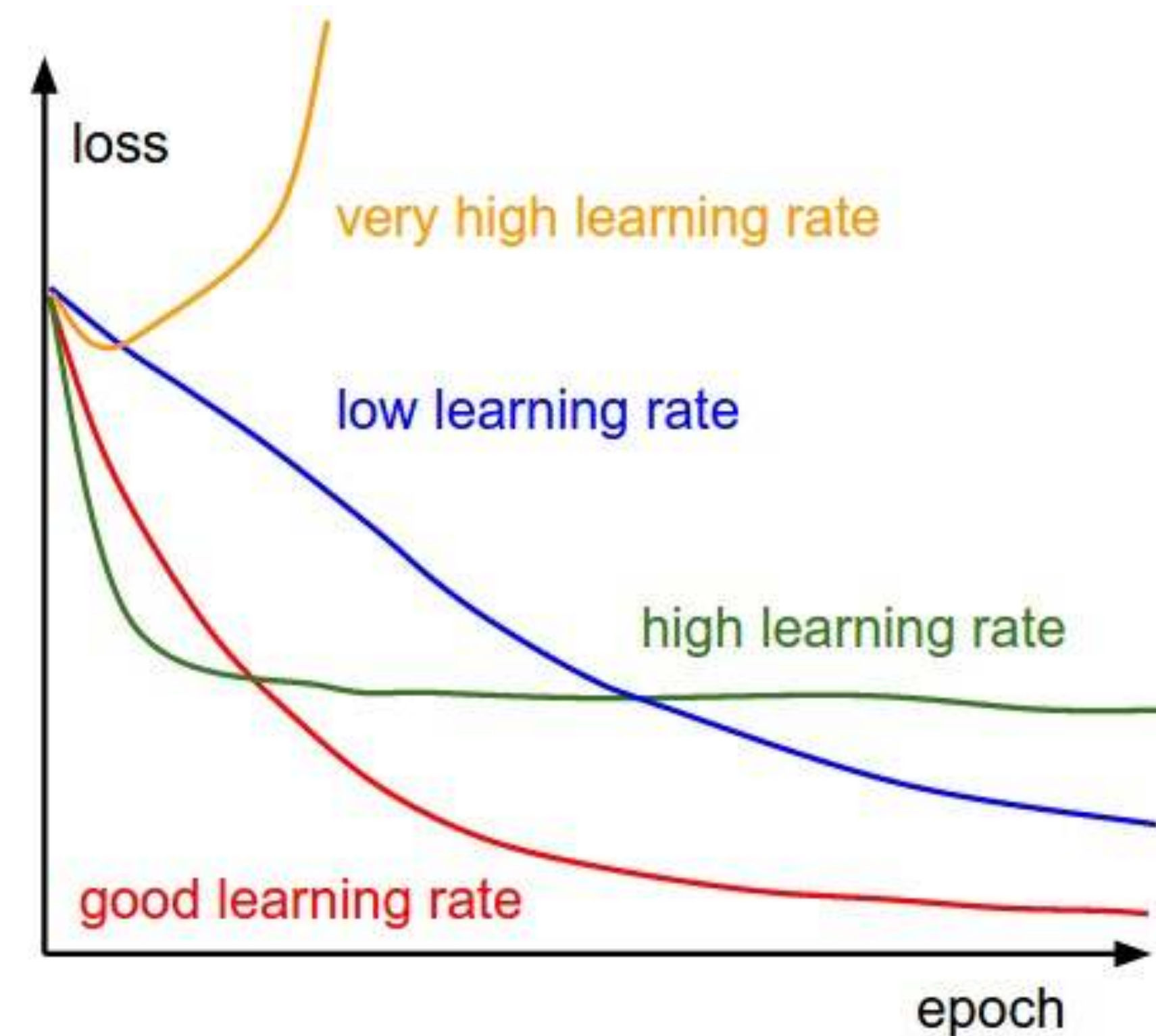
MULTI-LAYER PERCEPTRON

Multi-layer Perceptron [Hopfield, 1982]



MULTI-LAYER PERCEPTRON

- Solving a non-convex optimization.
- Iterative solution.
- Depends on the initialization.
- Convergence to a local optima.
- Judicious choice of learning rate



THANK YOU

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